

AKSHAY S

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SYSTEMS RESEARCH & TECHNICAL WRITING

Independent Systems Research

2025 – Present

Power delivery, ambient computing, and spatial hardware

- **The Architecture of Abundance: How Energy Flows and Creates Intelligence.** Worked through the physics of terrestrial power beaming under the 10 W/m² FCC limit and of space-based beaming, modeling inverse-square loss, aperture diffraction, and rectenna conversion ceilings.
- Calculated that removing the battery and onboard compute strips roughly 30% of a humanoid robot's power requirement, and argued for continuous ambient harvesting paired with supercapacitors to cover peak movement draw.
- **Computing Leaves the Device: Ambient Power, Distributed Intelligence, and the Spatial Internet.** Mapped the hardware requirements for sub-100 mW battery-free headsets that push rendering to nearby edge nodes inside a 20 ms motion-to-photon budget.
- Compared sending raw geometry against sending compressed scene descriptions, priced by what each costs the device in real-time power.
- **Ambient-Powered Computing: Why the End of the Battery Is a Systems Problem, Not a Power Problem.** Position paper on replacing chemical batteries with ambient RF power delivery across indoor spaces.

FOUNDING & OPERATOR EXPERIENCE

Growth & Marketing Lead

Feb 2025 – Apr 2025

Klarvia — B2B Corporate Wellness

- Wrote the first marketing narrative and pitch materials aimed at enterprise HR and corporate wellness buyers.
- Ran direct outreach to prospective corporate clients and read through term sheets during early fundraising conversations.

Founder & Prototyping Projects

2025 – 2026

Early-Stage Explorations

- **Wireless Power Architecture:** Spent a year on the hardware trade-offs behind wide-area wireless power transmission and pitched the architecture directly to venture funds.
- **Restaurant Pre-Ordering:** Built an MVP web app and interviewed local restaurant owners to test whether pre-ordering actually saved walk-in diners measurable time.
- **Local Wi-Fi Network:** Worked out deployment unit economics and business models for localized public Wi-Fi in high-footfall cafes.

SELECTED TECHNICAL WORK

Visa Approval Classification Pipeline | *Python, bagging & boosting models*

- Built a machine learning pipeline over roughly 776,000 records to predict visa approval outcomes.

Equity Market Cluster Analysis | *K-means, hierarchical clustering*

- Clustered NYSE stock data to examine how much diversification a portfolio was actually getting.

EDUCATION & LEADERSHIP

Indian Institute of Technology (IIT) Madras

2020 – 2025

B.Tech, Chemical Engineering

- Teaching Assistant and mentor in the Saathi Wellness Program, working with junior students on coursework and study habits.
- Table Tennis: 2x Gold Medalist (2022, 2023) and Vice-Captain, IIT Madras team.